

Bushel & Bone

A Dark Alt-1800s Survival & Management Game

Stardew Valley × Oregon Trail × Ethical Horror

Game Design Document

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1 Executive Summary

Bushel & Bone is a dark, alt-1800s survival and management game with Weird West undertow. The player runs a rooted homestead worked by farmhand clones grown by strange pseudo-science, buying and selling crops in a volatile regional market while managing supplies, labor, and the slow accumulation of moral debt. Cruelty toward the clones raises yields — and calls down a supernatural reckoning that the land remembers across generations.

The game plays in nested loops. Each Year is a tactical run in which the player must survive four seasons to reach spring. Each Farmer's Lifetime is an act — several years chained together, ending in death. Each Family Lineage on a plot of land is the true roguelite run, ending only when the family line dies out or the land itself is lost. Death of a farmer passes the farm to an heir; death of the line ends the campaign and grants meta-progression that carries into a new run on new land, in the same evolving world.

Gameplay pillars are planting, buying, management, supplies, and market — all treated as first-class systems. The tone is grit and survival: the joy is barely making it. Story is present but light, delivered through a small cast of deep NPCs in the town of Marrow's Cross, one town-wide Season Arc per year, and individual NPC threads that unfold across generations of play.

Bushel & Bone is designed mobile-first, portrait-primary, entirely turn-based. It ships as a Next.js web application deployed on Vercel, wrapped for iOS and Android via Capacitor for App Store and Play Store distribution. The backend is Neon (serverless Postgres) with Better Auth for authentication.

1.1 Elevator Pitch

You inherit a plot of land and a promise. Grow crops. Buy clones. Survive the winter. The land keeps a ledger of every kindness and every cruelty you spend on the vessels that work it — and when the ledger is heavy, the land collects.

1.2 Reference Games

- Stardew Valley — the rooted-farm feel, the town of neighbors, the seasonal cadence.
- Oregon Trail — the weight of menu-driven decisions, the resource-management-before-winter pressure, the honesty about death.
- Sunless Sea / Fallen London — the choice-card resolution grammar, the dark literary tone.
- Frostpunk — the moral cost of doing what must be done to survive.
- Rogue Legacy — the generational meta-progression, the "next heir inherits" beat.
- Balatro — the mobile-native session lengths and the clean roguelite loop shape.

2 Vision & Design Pillars

2.1 Core Player Fantasy

The player should feel, at the end of a session, that they survived another year — grit, scarcity, hard-won victories against nature. The emotional tone is closer to True Grit than to Stardew Valley: warm but not cozy, hopeful but never soft. The joy is in barely making it, not in optimizing yields. A run that ends with three graves marked, seventeen unmarked, and the family still eating in the spring is a triumph.

2.2 The Five Gameplay Pillars

All five are first-class systems, given equal design weight. No pillar is a garnish for another.

- **Planting** — crop selection, seasonal fit, field condition, harvest timing.
- **Buying** — supplies, tools, buildings, livestock, clones. The stock-up rhythm.
- **Management** — clone assignment, treatment, roster scaling, oversight structure.
- **Supplies** — coal, salt, seed, feed, medicine, ammunition, comfort goods. Oregon Trail's heartbeat.
- **Market** — volatile prices with seasonal patterns, demand shocks, information asymmetry, forward contracts.

2.3 Non-Goals

- No real-time action or dexterity mechanics. The game is fully turn-based.
- No walkable character on a Stardew-style tile grid. The Homestead View is a diorama.
- No dialogue trees beyond 2–4 choice menus per exchange.
- No multiplayer at MVP.
- No pay-to-win mechanics under any monetization model considered.
- No depictions of real Native American cultures. See §3.4.

3 Setting & Tone

3.1 Setting: Alternate 1800s Weird West

Bushel & Bone is set in an invented alt-1800s frontier — not a historical period-piece, but a version of the 1800s in which the technology and the supernatural do not obey the rules of our timeline. Wagons, general stores, telegraph, rifles, and rail all exist. So do bodies grown in glass tanks and things that walk at night after dishonored burials. The specifics of geography, nations, and history are invented so we have freedom to shape events without breaking real-world lore.

3.2 The Weirdness Dial: Weird West / Occult Undertow

Something is genuinely wrong out there. The supernatural is real, present, and slowly revealing. It is not fantasy — there are no wizards, no magic swords, no dragons. It is horror-adjacent: cults in the hills, cryptids in the woods, wells that go too deep, the dead who do not stay buried when they were wronged. Tonally: Blood Meridian, Deadlands, Sunless Sea, The Witch, Bloodborne's pastoral early hours.

3.3 Tone Guidelines

- Grounded first. Weird second. The player spends most of their time worrying about wheat and coal, not spirits.
- The supernatural is slow to reveal. Whispers before Walkers. Ambient dread before overt horror.
- Restraint over spectacle. A single crow at the window means more than an army of ghosts.
- Adult tone. Not gratuitous, but not softened. Death is real. Cruelty has weight. Kindness is scarce and precious.
- No fourth-wall gags, no memes, no anachronistic humor. This is a game with a face.

3.4 Cultural Care Note

The 1800s American frontier setting invites certain traps. Bushel & Bone deliberately avoids them. No Native American cultures — real or thinly-veiled — are depicted. If any "old people of the land" appear in the game world (e.g., a folk-magic community adjacent to Old Nan), they are clearly invented and treated with dignity. "Bandits" and other hostile outsiders are individuals or ideologically-defined groups, never ethnic ones. This is a design constraint, not a suggestion.

4 The Core Hook: Farmhand Clones & The Reckoning

4.1 The Central Premise

The player's labor force is not human hires. It is clones — adult bodies grown by strange 1800s pseudo-science, sold from wagons and cultivated in glass tanks. This is the game's single most distinguishing feature and the axis on which its moral system turns. Every farmhand is a decision the player made.

4.2 The Moral Axis

The player can treat clones with kindness or cruelty. Cruelty produces higher yields in the short term — more work extracted per body, less coin spent on comfort, more crops harvested per season. Kindness produces slower results but sustainable loyalty, higher household morale, and a lower risk of catastrophe.

The choice is present in nearly every daily decision. Assignment options include cruel modifiers ("double shift," "no rations today"). Housing choices affect clone health. Burial choices affect the ledger. Even the sourcing of clones (buy from the corporate merchant, grow in the vat, take in a foundling) is a moral posture.

4.3 Consequences: The Land Remembers

The world pushes back through supernatural means, not social or economic ones. Cruelty accumulates in a hidden Reckoning Meter that manifests, at rising thresholds, as escalating horror: cursed harvests, restless dead, walkers in the night, and eventually a Reckoning that comes for the farmer directly. The system is designed to feel like the land itself has memory — poetic, un-gameable, and thematically the tightest possible expression of the pitch.

Full detail on the Reckoning tier system is in §14.3.

5 Structure: The Nested Run Architecture

Bushel & Bone uses a three-tier nested run architecture. This is the roguelite spine — the single most important structural decision in the design. Every gameplay system is scoped to one of the three tiers.

5.1 The Year — The Tactical Unit

Every year runs Spring → Summer → Fall → Winter (four seasons of twenty days each; 80 days total). Winter is the moment of judgment: did the player stockpile enough food, coin, fuel, and living clones to make it through?

- **Success:** the farm earns a permanent improvement — a new field cleared, a building added, a reputation notch, a "Year Survived" marker etched somewhere visible on the homestead.
- **Failure:** clones die of exposure, coin is lost paying debts, land improvements decay, and the farmer may take a mortal wound or age catastrophically. Painful, but not final.

5.2 The Farmer's Lifetime — The Act

The active farmer ages a year every in-game year. After a certain age, natural death chance climbs. Farmers can also die from cruelty reckoning, disasters, accidents, or an especially catastrophic winter. Farmers accumulate personal traits over their lifetime — a scar from a bad winter, a superstition after a haunting, an heirloom rifle passed down. When they die, a beat of reflection follows: a funeral, a grave placed on the property, the estate passes.

5.3 The Family Lineage — The True Run

The real run is measured in the land and the line. A successor picks up on the same soil, inheriting the farm's buildings, its graves, its reputation, its curses, and a portion of its coin. The land itself accumulates history — this corner is where three clones died in '69, that corner grew the record harvest in '82, this well went bad after the reckoning of '77.

5.4 Run-End Conditions

A campaign ends only when one of two things happens:

- **The line dies out.** No successor available and no coin to hire one from town.
- **The land is lost.** Foreclosed by the bank, cursed past use, or burned to ash.

At that point, the Legacy Report fires (§13.1) and the player earns Vigils toward meta-progression on a new patch of land.

6 Time & Pacing

6.1 Turn-Based Days

Time advances only when the player taps "Advance Day." The day is the atomic time unit. This makes the game save-anywhere trivial, mobile-native by default (playable in line at a coffee shop, resumable a week later), battery-efficient, and honest to its Oregon Trail lineage.

6.2 The Calendar

Unit	Length
Day	Atomic — one Morning Brief, one Assignment, one Play beat, one Dusk Report
Season	20 days
Year	80 days (Spring 1-20, Summer 21-40, Fall 41-60, Winter 61-80)
Farmer's Lifetime	5-20 in-game years (typical), 3-10 hours active play
Family Lineage	Unbounded — a run continues until line-death or land-loss

6.3 Session Length Targets

Session Shape	Duration & Content
Snackable	2-4 minutes — a few days of check-in, dismiss an event, read the market
Standard	10-15 minutes — a full season or half a year
Long	30-45 minutes — a full year, planting through winter
Deep Dive	1 hour+ — a farmer's peak years, multiple seasons at a stretch

6.4 Day/Night Sub-Beats

Each day advances through four visible sub-beats. The player does not play each sub-beat separately; they witness them as art and lighting transitions.

Beat	What Happens
Dawn	Morning Brief fires. Weather revealed. Assignment happens. Some events (a stranger at the gate) are dawn-specific.
Midday	Town is open. Market prices post. Most work happens. Positive random events skew midday.
Dusk	Work ends. Meals. Social events. Wildlife events skew dusk (the wolf comes).
Night	Reckoning events fire almost exclusively at night. Weather intensifies. Sleep events

	(dreams, visitations) live here.
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Moon phase matters. Full moon nights are rarer and wilder — moon barley grows more, walkers walk more, Whisper Wheat is loudest. New moon nights are quieter but reckoning events that fire are darker.

7 Core Screen & Day Structure

7.1 The Homestead View

A 2D scene of the farm, framed like a diorama — a tableau, not a walkable map. Fields on the left, farmhouse in the middle, barn/pen on the right, the road to town in the distance. Weather, lighting, and season shift the whole image. In winter, snow covers the fields. After the reckoning, the crows come.

7.2 HUD & Tab Bar

Persistent HUD around the edges of the Homestead View:

- Top: current date, season, weather forecast, moon phase.
- Bottom left: coin, household morale, farm reputation.
- Bottom right: the day's action button ("Advance to Dusk" / "End Day").
- Tap the scene to zoom into a field, a clone, a building — inspect, assign, or manage.

The bottom tab bar is the game's navigation spine:

- Homestead — this scene, the default view.
- Clones — roster, health, mood, cruelty ledger.
- Fields & Stock — crops, animals, storage, tools.
- Market — prices, buy/sell, contracts.
- Town — map of destinations, NPCs, quests (locked until you travel).
- Ledger — history, family tree, graves, Codex, records, reckoning insight.

7.3 The Three-Beat Day

Every day has a fixed rhythm. Not every beat needs input, but they always exist.

1. Morning Brief (Dawn)

A pop-up card with the date and weather, any overnight events, and any decisions pending. Total time: 5-20 seconds.

2. Assignment (still Dawn)

The Clones tab briefly foregrounds. The player drags each clone to a task: plow, plant, harvest, water, forage, repair, rest, chore duty. Cruel options exist here too — "double shift," "no rations today" — with visible trade-offs. If assignments aren't changed, they keep yesterday's. Total time: 0-30 seconds.

3. The Day Plays Out (Midday → Dusk)

Tap "Advance." The scene animates through midday and dusk in 2-5 seconds. If nothing happens, the day lands at Dusk Report. If an event fires, the game pauses on that beat and shows a choice card.

4. Dusk Report (Night)

A ledger-like summary of the day: crops advanced, coins earned, morale changes, injuries, deaths, market movement. Any night-triggered supernatural activity happens here.

7.4 A Day in Practice

- **Quiet day in early summer:** tap "Advance," watch the sun move, land at Dusk Report showing "Wheat 40%, Corn 60%, +2 coin from odd jobs." Ten seconds, next day.
- **Market day:** Morning Brief mentions "Grain buyer in town today, prices up." Tap into Market, sell 40 bushels at a peak, advance. Forty seconds.
- **Reckoning day:** Morning Brief is silent. Assignment is normal. At Dusk, a card interrupts: "Silas' grave is disturbed. The fence is broken outward. Something walked." Choose to investigate (dangerous), bar the doors (safe but suspicious), or ignore. The choice ripples for weeks.

8 Farming Systems

8.1 Crop Families

Four families, each serving a distinct economic and thematic function.

Grains — the staple. Low margin, steady demand, defensive.

- **Wheat** — 30-day grow, Spring→Summer, high volume when it hits, bread money. The safe bet.
- **Corn** — 25-day grow, Summer, versatile (sell, feed animals, brew whiskey).
- **Oats** — 18-day grow, cool-season, shoulder crop. Mostly fodder.

Roots — winter survival food.

- **Potatoes** — 22 days, store through winter, low sell price but life-saving.
- **Turnips** — 15 days, cool-season, bulk food, minimal coin.

Cash crops — the swings. High margin, high labor, market-dependent.

- **Tobacco** — 40 days, Summer, very high price when demand spikes.
- **Cotton** — 45 days, Summer, requires many hands to harvest. Massive payoff if properly staffed.
- **Hops** — 30 days, Summer, brewery demand, wild price swings.

Weird crops — the tempting ones. Locked until discovered.

This is where farming meets the reckoning system directly.

- **Moon Barley** — 20 days, only grows if planted under a full moon. Unpredictable yield but sells for silver.
- **Bone-root** — 25 days, grows better in fields where a clone has died. Fetches occult prices in certain markets.
- **Whisper Wheat** — 30 days, if left unharvested at night, whispers to nearby clones (passive morale effects, some positive, some ruinous). The rumor is it's growing from something underneath.

8.2 Field Mechanics

- **Fields, not tiles.** The player owns N fields (start with 2-3 cleared). Each field holds one crop type at a time.
- **Field size:** Small (4-bushel potential), Medium (8), Large (16). Bigger fields need more labor.
- **Field condition:** Fertility (drops after harvest, restored by fallow/manure/rotation), Moisture (weather-driven), Taint (rises with clone deaths and dark rituals — enables bone-root, repels normal crops).
- **Field expansion:** Clearing a new field is a multi-day labor project requiring tools and coin.

- **Named fields:** Every field has a name and a history. "West Field, where Silas fell in the winter of '68." Graves live on specific fields.

8.3 Growth Mechanics

Every planted field shows a 0-100% progress bar. Per-day growth is a product of base speed, seasonal fit, weather, field condition, tending (a farmhand assigned to "tend" adds a per-day bonus), and any reckoning modifier.

When a crop hits 100%, it enters a 5-10 day harvest window. Miss it and the crop rots or is destroyed by weather.

8.4 Seasonal Planting Windows

Season	What Happens
Spring (Days 1-20)	Plant most grains, roots, and early cash crops.
Summer (Days 21-40)	Harvest early grains. Plant late cash crops. Tend hard.
Fall (Days 41-60)	Main harvest. Last-chance turnips. Prep for winter.
Winter (Days 61-80)	Nothing grows outdoors. Market and reckoning games dominate. Late runs can unlock a Glasshouse for indoor growing at fuel cost.

9 Farmhand Clones — The Labor, The Moral Engine

9.1 The Three Sources

1. The Clone Merchant (Available from Day One)

Dr. Ambrose Vane, a traveling doctor-showman, rolls into Marrow's Cross every 15–30 days with a wagon of "new-blooms" for sale. He sells healthy adult clones at premium prices. Corporate, clean, priced like livestock. The moral weight is complicity in the trade — buying from him supports whatever mill produced these bodies.

2. The Vat (Build Mid-Run)

A large glass tank installed in the barn. The player feeds it material and, over 15–20 days, it produces a clone from a seed. Cheaper per unit than the Merchant, but risky: malformations are common, some batches die in the tank (each death a small reckoning tick), successful ones are unpredictable. Grow-your-own DIY horror. When townsfolk see the Vat, they judge you.

3. Foundlings (Rare Random Events)

Sometimes a clone shows up at the gate — escaped from another farm, half-starved, of unknown quality and history. The player chooses whether to take them in, turn them away, or turn them over to the sheriff for a bounty. Free labor with a mystery attached.

9.2 Anatomy of a Clone

- **Name.** Auto-generated with alt-1800s flavor (Silas, Elias, Ambrose, Josephine, Meredith). Player can rename. Naming matters — the XCOM effect.
- **Age.** In "seasons of service." Clones age faster than humans. Peak around 4–6 years, decline after 8, burnout by 10–12 barring accident.
- **Body (weak/average/strong/powerful).** Heavy labor throughput.
- **Mind (dull/average/sharp/keen).** Tending crops, market savvy, oversight capability.
- **Constitution (fragile/average/hardy).** Sickness and injury resistance.
- **Traits (1–3 quirks).** Examples: superstitious, kind-hearted, defiant, deft-handed, sleepwalker, blessed, sees things, hollow.
- **Mood.** Very low → very high. Driven by treatment, food, rest, workload, funerals attended, witnessed hauntings.
- **Loyalty.** Slow-building. Rises only with sustained kindness. Very loyal clones work harder in crisis.
- **Ledger (private).** Every act of cruelty received. The clone never forgets. Neither does the land.

9.3 Archetypes

- **Field Hand** — cheap generalist.
- **Plower** — strong body, good at heavy field prep.

- **Grower** — sharp mind, tending crops grows them faster.
- **Herder** — good with livestock.
- **Cook** — assigned to the kitchen, boosts morale via meals.
- **Overseer** — rare, keen-minded; can supervise others, removing the need to hand-assign. Late-game scaling relief.
- **Malformed** — cheap, sad, short life. Often has a rare weird trait (talks to Whisper Wheat, sees hauntings before they land).

9.4 The Four-Axis Cruelty Ledger

Cruelty is not a single meter. It manifests across four channels with different tempos and different visibility.

- **Reckoning Meter (hidden).** The supernatural accumulator. Rises with clone deaths under cruel conditions, whip use, unfed workdays, dishonored burials. When it crosses thresholds, the world responds (§14.3). Visible only in retrospect until it is very high.
- **Household Morale (visible).** Aggregate clone mood. Cruelty tanks it. Low morale = slow work, more sickness, occasional refusals.
- **Town Reputation (visible).** NPCs notice patterns. A cruel farmer gets worse market prices, refused by the preacher, watched by the sheriff.
- **Ghost Roll (hidden).** When a clone dies wronged, they enter a private list. The reckoning draws from this list when it walks. Kindness at burial removes names from the roll.

9.5 Death & Burial

Every dead clone becomes a choice:

- **Bury on the land, marked grave.** Costs coin (marker, rites). Reduces reckoning. Grave stays visible on the Homestead View. Enables bone-root at cost.
- **Bury unmarked, in the field.** Free. No reckoning reduction. Fully enables bone-root. Ghost Roll ticks up hard.
- **Sell the body to Dr. Vane.** Coin for the corpse. Reckoning ticks. Household morale drops.
- **Feed to the Vat.** Free "material." Massive reckoning tick. The next batch is strange.
- **Give a full funeral.** Costs meaningful coin, takes half a day of labor, all clones attend. Big reckoning reduction. Big morale boost. Establishes you publicly as that kind of farm.

9.6 Roster Scaling

Housing Tier	Clone Cap
Starting (Farmhouse only)	1
Bunkhouse (early)	4
Barracks (mid)	8

Great Bunkhouse (late)	15
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Higher counts require more food, more oversight, more housing upkeep. More is not always better. Twelve unhappy clones will destroy a farm faster than four content ones.

10 Buildings, Tools, Livestock, Supplies

10.1 Buildings

The 2D diorama fills in over time. Every building is visible on the scene.

Starting: Farmhouse, small Barn, one Storehouse.

Early (Year 1-2, cheap)

- Bunkhouse — raises clone cap to 4.
- Well — reduces drought vulnerability.
- Chicken Coop — unlocks chickens.
- Additional Storehouse — raises stockpile cap.

Mid (Year 3-5, meaningful investment)

- Barracks — clone cap 8.
- Kitchen — unlocks Cook role; better meals = better morale.
- Smokehouse — preserve meat through winter.
- Distillery — corn/hops → whiskey (huge margin, town scrutiny).
- The Vat — clone growing.
- Family Plot — dedicated burial ground; funeral rites here reduce reckoning more effectively.

Late (Year 6+, transformative)

- Great Bunkhouse — clone cap 15.
- Glasshouse — grow one crop through winter, at fuel cost.
- Cellar — long-term storage, prices held for better market timing.
- Shrine — dedicated to a saint/spirit; small ongoing morale + reckoning reduction.
- Ossuary — a dark inverse of the Shrine; keeps clone bones on site with mechanical benefits and terrible costs.

Every building requires coin + lumber + nails + clone-days, plus a season's fair weather to raise.

10.2 Tools

Throughput multipliers purchased in town. Each permanently improves one kind of action.

- Iron plow → Steel plow → Steam plow (removes need for strong-body clone in field prep).
- Scythe → Reaping machine (faster grain harvest, wider harvest window).
- Threshing machine (cuts post-harvest grain labor days).
- Grain elevator (reduces spoilage in storehouses).

- Rifle (defense against wildlife, bandits, worse).
- Wagon upgrades (faster town trips, larger carry capacity).

10.3 Livestock

- Chickens — cheap, eggs daily, low upkeep. Foxes love them.
- Pigs — convert food scraps to meat and lard. Escape prone.
- Cows — milk, calves. Big investment. Vulnerable to sickness.
- Sheep — wool. Slow but reliable.
- Draft horses — labor multiplier; assign to a plow team and one clone does two clones' work.
- Weird livestock (rare unlocks) — hollow-eyed cow whose milk never sours but rots the pail; ghost hen that lays a silver egg once per season.

10.4 Supplies — The Oregon Trail Buy-List

- Seed — per crop, per planting. No seed = no planting.
- Feed — for livestock; skimp and animals die.
- **Coal / Firewood** — winter fuel. The winter survival check. Underestimate and clones freeze.
- Salt — food preservation.
- Lumber & Nails — building and repairs.
- Whiskey / Tobacco / Sugar — comfort goods; small ongoing morale.
- Medicine — treat sickness; expensive, sometimes out of stock.
- Ammunition — wildlife, bandits, worse.

11 The Market Economy

The market is the game's second heartbeat. It has to reward information, punish greed, and never feel like pure RNG.

11.1 Four-Layer Price Model

- **Layer 1 — Base price.** Inherent value. Static.
- **Layer 2 — Seasonal wave.** Every commodity has a predictable seasonal curve. Grain cheap after fall harvest, expensive by late spring. Players learn these curves.
- **Layer 3 — Demand shocks.** Random-but-telegraphed events that spike or crash a commodity for a bounded period.
- **Layer 4 — Micro-noise.** Small daily variance ($\pm 3\text{--}5\%$). "Sell at exactly the peak" is impossible.

Player transactions have minor market impact. Dumping 200 bushels sags local price for a week — encourages spreading sales.

11.2 Information Sources

- Local Market prices — free, always visible.
- The Ledger's price history — last 30 days for anything ever transacted.
- **The Weekly Broadsheet** (bought in town). Rumors with confidence tags (Rumored / Reported / Confirmed). Some are planted misinformation.
- **The Telegraph Office** (mid-game). Real-time prices from distant markets. Subscription per season.
- Traveling merchants and NPCs — some townsfolk drop information as dialogue if the relationship is warm.
- **The Almanac** (rare buy). Seasonal projection of expected demand, with predicted moon phases and weather trends.

11.3 The Four Venues

- **1. Local Town Market.** Always open. Small volume (5-20 bushels per trip). Reliable, boring, safe.
- **2. Regional Buyers.** Named merchants who arrive with take-it-or-leave-it deals. Reading the buyer is the game.
- **3. The Rail Depot** (mid-game unlock, day's wagon ride away). Ships to distant cities. Highest prices for grain and cotton, but shipping cost and 5-10 day delay.
- **4. The Black Market** (unlock via reputation drop or dark contact). Only venue that buys bone-root, whisper wheat, and clone bodies at fair value. Being seen there tanks town reputation.

11.4 Contracts — The Forward Game

Beyond spot sales, merchants offer forward contracts.

- **Standard sell forward.** "40 bushels of wheat, delivery Day 45 of Summer, \$9/bu." Guaranteed price if delivered; penalty in coin and reputation if not.
- **Speculative buy forward.** Lock in supply costs. Bet on future prices.
- **Quality-locked.** Contracts require specific grades or origin ("virgin field," "no touched clones"). Weird West flavor.
- **Company standing orders** (late-game). A distant industrial buyer commits to buying everything at a fixed rate.

Contracts create pressure. Committed to deliver 60 bushels by Day 45? A hailstorm on Day 30 is a genuine crisis. This is where farming and market weld together.

11.5 Buying Supplies

The same layered price model applies to buying. Feed is cheap in summer, expensive in winter. Coal spikes when a mine collapses upriver. The stock-up rhythm — heavy in fall for winter, spending down in spring — becomes the year's heartbeat.

11.6 Long-Term Trends

Over multiple in-game years, some commodities drift. Cotton might slowly rise over a decade as textile mills industrialize. Whiskey might crash after a temperance movement. These are generational trades — planting an orchard in Year 3 that pays off in Year 8 is the long game. Multi-generational campaigns exploit this deeply.

12 The Town of Marrow's Cross

The town is a small rail whistle-stop that never quite grew. Bones under it. A cross above it. It is where information, supplies, contracts, moral judgment, and story converge.

12.1 Travel Model

Going to town is a one-day round trip. Tap "Travel to Marrow's Cross" from the Homestead View, the day advances, and the player lands on the Town View. All town business is done in one visit. The day is gone; farm assignments ran on template. Wagon upgrades shorten this. Certain townsfolk occasionally come to the farm, sparing the trip.

12.2 Town View

A 2D diorama of a dusty crossroads with the buildings visible. Tap a building for its menu screen. NPCs are visible on the scene when present; tap a silhouette to talk.

12.3 Venues

MVP Core (Available From Day One)

- General Store — supplies, tools, small livestock.
- Grain Exchange — the local market; buy/sell all crops.
- Doctor's Office — medicine, treatment, rumors.
- The Saloon — meet NPCs, hire foundlings, hear things, buy comfort goods.
- The Church of the Long Vigil — funerals, blessings, moral pressure.
- The Bank — savings, small loans, holds mortgages.
- The Halloway Broadsheet — buy the weekly paper, post ads.
- The Boarding House — heirs available for hire, marriages.

Post-MVP / Mid-Game Unlocks

- Telegraph Office — subscribe for distant market prices.
- Sheriff's Office — cruelty investigations, foundling bounties.
- The Undertaker — professional burial services.
- Dr. Vane's Wagon (transient) — clone merchant.
- The Old Well (narrative unlock) — the town's Weird West secret.

12.4 The NPC Roster

Ten named NPCs. Each has a relationship track (Cold / Neutral / Warm / Bonded), a personal story arc across many in-game years, and specific opinions about the cruelty ledger. Small cast, deep characters.

- **Mayor Cyrus Halloway.** The town's political center. Offers civic contracts and unpleasant favors. Warmth = access to town-wide policies.

- **Preacher Elias Grange.** The moral eye. Blesses your farm publicly if kind; condemns from the pulpit if cruel.
- **Doc Bell.** Cynical, sees everything, sells medicine and rumors. Medicine gets cheaper the more clones you send him alive.
- **Meredith Vane.** The saloonkeeper. Hears everything. Related to Dr. Vane (his sister; see §12.5). Warmth unlocks trade tips and Foundling introductions.
- **Silas Ridley.** The banker. Cold, correct, holds mortgages. Foreclosure is his tool.
- **Bess Halloway.** The mayor's daughter. Potential spouse for the farmer — cleanest path to a lineage heir.
- **Old Nan.** Folk-magic woman on the edge of town. Knows the reckoning is real. Sells nothing but tells things others won't.
- **Sheriff Nathaniel Coldwater.** The law. Investigates cruelty complaints. Enough investigations = fines, seizures, charges.
- **Dr. Ambrose Vane.** The clone merchant. Arrives every 15–30 days. Warmth = better selection, discounts, occasionally a rare clone.
- **Sister Ruth Grange.** The preacher's wife. Runs charity work. Donations here boost morale, reduce reckoning, unlock a "kind farm" reputation track.

12.5 The Vane Siblings — Deep Story Hook

Meredith Vane and Dr. Ambrose Vane are siblings by blood. Ambrose is the elder — a natural philosopher trained back east who came to the frontier because Marrow's Cross does not ask questions and the rail brings customers to him.

Meredith is fifteen years younger. She grew up in his shadow, cleaning up after him, learning enough of the process to be dangerous. She left at nineteen after a batch went wrong in a way she could not unsee. She landed in Marrow's Cross a decade before the player arrives, bought the saloon with money nobody asks about, and has never turned him in. She also has never once eaten a meal at his table.

The town knows they are siblings. The town does not know why she stays close. Three possible truths — randomized per campaign — surface across deep NPC threads:

- **1. She is watching him for someone else.** A rival scientist, a religious order, a shadow authority. She reports quarterly.
- **2. They made something together that neither can undo.** A third Vane, alive somewhere. Neither will move first.
- **3. She is one of his.** His first successful adult vessel, imprinted with his real sister's memories after the real Meredith died in childhood. She suspects. He will not confirm.

Which truth this run holds is set at campaign start. Uncovering it takes deep warmth with Meredith and, in some cases, warmth with Ambrose at moral cost.

12.6 Interaction Model

Conversations are 2–4 choice menus per exchange — Reigns-level economy of words. Every NPC has a memory. Reputation is per-NPC plus overall town reputation.

12.7 Season Arcs & NPC Threads

The light story lives at two scales:

- **Season Arcs.** One town-wide story per year, unfolding across ~40 in-game days. Examples: Year 1 "The Newcomer" (a new family arrives, tension over land); Year 2 "The Preacher's Sickness"; Year 3 "The Rail Comes"; Year 4 "The Vigil Breaks" (reckoning-tied). Arcs react to player state.
- **NPC threads.** Personal quests each named NPC offers over time on their own tempo. Bess courts (or is courted). Doc Bell has an addiction. Sheriff Coldwater investigates a stranger.

12.8 Festivals

- Winter's End Fair (early spring) — new seeds, contracts posted, planting kicks off.
- The Long Vigil (midsummer) — night of prayer at the church. Attending reduces reckoning. Skipping is noticed.
- Harvest Festival (fall) — premium prices, morale boost, marriages happen here.
- Ash Night (deep winter) — the darkest day. Farms are haunted more. The reckoning walks.

Festivals are soft-required attendance. Skipping the wrong one has consequences. Attending the right one at the right moment can save a run.

13 Roguelite Meta-Progression

13.1 Run-End: The Legacy Report

When the line dies out or the land is lost, the game slows down. The Legacy Report is a sequence, not a stat screen:

- The Family Tree unfurls — every farmer who worked this land, names, years, causes of death.
- The Ledger reads itself — best harvest, worst winter, total coin, clones bought, clones buried, cruelty peak, kindness peak.
- The Graves speak — each clone buried with a marker gets a single line rendered.
- The Ending — a procedurally-assembled paragraph from the run's actual events.
- Vigils earned.

Emotionally, this should feel like the closing pages of a novel — two or three minutes of real reading. Skippable, but rewarded for lingering.

13.2 Vigils — The Meta-Currency

Vigils are earned at run's end based on years survived, longest lineage, clones honored (funerals, marked graves, low Ghost Roll), story arcs completed, records set, and cruelty completions (grim path also earns Vigils — a full black-market lineage that survives is its own achievement).

13.3 The Unlock Catalog

Vigils buy new content and starting conditions, not power. New possibilities, not "you are stronger now."

Starting Archetypes

- The Homesteader (default first run).
- The Widow — starts with a grown heir, no clones, higher morale ceiling.
- The Disgraced Doctor — starts with a Vat built, town reputation Cold, cheaper clones.
- The Escaped Prisoner — starts with \$0 and a watching Sheriff, but high physical stats.
- The Preacher's Son — starts with a Shrine, blessed by Elias, forbidden certain profits.
- The Vane Cousin — family to them. Discount at Vane's Wagon. Meredith will not speak to you.

Other Unlock Categories

- New Crops — moon barley, bone-root, whisper wheat, and others unlock permanently once discovered.
- New Buildings — Ossuary, Distillery, Glasshouse, etc.
- New Season Arcs — additional yearly stories added to the pool.
- New NPCs — Marrow's Cross grows over campaigns.

- Marrow's Cross Landmarks — permanent town changes based on completed arcs.
- Codex Fragments — story lore, backstory, revealed truths.
- Ascension Modifiers — unlockable difficulty options.

13.4 The Codex — Persistent Memorial

The Ledger tab has a Codex section that persists across every run ever played:

- The Book of Farmers — every named farmer, every heir, years and death.
- The Book of Clones — every clone, cause of death, field worked, honor of burial.
- The Book of Years — every year survived, with weather and notable event.
- The Book of Records — personal bests. Longest lineage, highest yield, coldest winter survived.
- The Book of Truths — story lore uncovered.

This is the game's true persistence. It is what a player shows a friend after a year of playing.

13.5 The New Land — Starting Fresh, Same World

A new campaign starts on a new patch of land in the surrounding valley. Marrow's Cross remains Marrow's Cross. Land region options unlock over time:

- Riverbend Farm — fertile, flood-prone.
- Highwater Ridge — thin soil, harsh winters, cheap land, best for survival challenge.
- Coalridge Bottom — near a mine, extra coin from side hustle, bad air.
- Widow's Cross — cursed land, cheap purchase, reckoning starts at 20% baseline.
- The Meadows — beginner-friendly, low risk / low ceiling.
- The Vane Homestead (rare late unlock) — the actual former home of Ambrose. Fully haunted.

13.6 Marrow's Cross Evolution

Buildings paid for or resolved into existence in past campaigns are still there in the next one. NPCs influenced have new dialogue. A preacher helped might have a statue in the square. This makes each new campaign feel like a new chapter of the same book, not a factory reset.

13.7 Ascension Modifiers

Optional, stackable difficulty modifiers. Each one adds a challenge and increases Vigils earned.

- +1 The Cold Bites — winters 25% harsher.
- +2 The Market Moves Fast — demand shocks twice as volatile.
- +3 The Land Remembers — reckoning accumulates 25% faster.
- +4 No Foundlings — no free clones. Merchant only.

- +5 The First is Wronged — start with one clone whose Ledger is already stained.
- +6 The Bank Watches — Silas Ridley offers no loans.
- +7 Silent Nan — Old Nan will not speak until Year 5.
- +8 The Broadsheet Lies — half of newspaper information is false.
- +9 One Season Shorter — the year is 60 days, not 80.
- +10 The Vessel is Weak — all clones age twice as fast.

13.8 What Carries vs. What Doesn't

Carries Between Runs	Does Not Carry
Vigils and unlocks	Coin
Codex (all books)	Buildings on old land
Unlocked archetypes, crops, buildings, arcs	Clones (they died with the land)
Marrow's Cross evolution	NPC relationships (fresh)
Ascension progress	Farm reputation

14 Random Events & Day/Night Systems

The event layer is where the game's personality lives. It is what makes each year a story worth telling.

14.1 The Seven Event Families

- **1. Weather** (mostly deterministic, seasonal, forecast-visible). Rain, drought, hail, frost, blizzards, dust storms, aurora over the north sky in deep winter. Weather compounds — three dry weeks trigger drought.
- **2. Pests & Disease** (probabilistic, escalating). Locusts, crop blight, fowl pox, cattle sickness, clone fever. Small outbreak ignored becomes large one next season.
- **3. Wildlife & Outsiders** (probabilistic, escalating). Foxes on chickens, wolves on sheep, bears rarely, bandits brutally. Guard clone on watch deters most.
- **4. Opportunities** (positive random events). Traveling salesman with rare seeds, wandering doctor with experimental medicine, foundling arrives, buried purse, old friend writes.
- **5. Town Events** (affect market, reputation). New buyer arrives, bank raises rates, preacher publishes sermon, fire on Main Street, election shifts policy.
- **6. Personal Events** (clone- and farmer-specific). Clone sickness, visions, love between clones, farmer marriage, births, deathbed choices, grief cascades.
- **7. Reckoning Events** (keyed to hidden Reckoning Meter). The world responding. See §14.3.

14.2 Event Resolution: The Choice Card

Every non-ambient event fires as a choice card — same visual grammar throughout the game. A framed illustration, a paragraph or two of prose, 2–4 choices with visible short-term costs and hidden long-term ripples.

Example — early-summer wolf event, low farm reputation:

A wolf came in the night. You find the sheep pen open, three sheep missing, the fence chewed and clawed. Rifle at your side, you scan the tree line. Your clone Silas is white-faced beside you.

- A. Track it. Silas comes with you. (Half-day of labor. May recover a sheep. Silas gets a scar and a story. Small injury risk.)
- B. Fortify the fence and set watch. (Full day of labor. Clone on watch every night for a week. No more losses if fence holds.)
- C. Do nothing. Take the loss. (No labor cost. Wolf returns. Livestock morale drops. Silas remembers you did nothing.)
- D. Set poisoned bait. (Small coin cost. Kills the wolf. Risks killing your own dog. Cruel-adjacent, small reckoning tick.)

Some cards are multi-day. Some are timed — respond within N days or the world resolves the event against you.

14.3 Reckoning Tiers

The Reckoning Meter is hidden. Its consequences escalate through five tiers.

- **Whispers (low).** Small ambient signs. Crows massing on the barn. Milk sours faster. A clone dreams of drowning.
- **Warnings (mid).** A dead crop row the day after a denied funeral. The well tastes of iron for a day. A clone refuses to enter a field.
- **Walkers (mid-high).** Dead clones return at night. Each walker is a specific dead clone with a specific grievance. Naming them at burial matters here.
- **The Long Vigil Fails (high).** The town's protective rite does not hold this year. Marrow's Cross itself feels it.
- **The Reckoning Proper (endgame).** A single great walking thing comes to collect. Can be resisted, appeased, or bargained with — but not ignored. Often ends a lineage.

14.4 Event Triggering Engine

Three overlapping systems roll events:

- **Calendar layer.** Festivals, seasonal weather, election days, fixed dates.
- **State layer.** Event pools filtered by farm state. A Vat opens whole event families. A haunted field triggers field-events. Low morale opens "clones consider running away."
- **Threshold layer.** Hidden meters cross thresholds and unlock specific escalations.

Event probabilities are tuned, not random. Locusts in high summer with dry weather = high probability. Locusts in winter = impossible. The world feels learnable even when it feels chaotic.

14.5 Event Pool Sizing

Milestone	Event Count
MVP launch	~120 unique events
Post-launch (6 months)	~250
Year one target	400+

Most events are 1-3 cards. Story-arc events chain 5-15 cards over many days. Content is written in lightweight event script (YAML/JSON) — no engineering required to add.

15 Art Direction: The Illustrated Almanac

The visual identity: an alt-1800s illustrated almanac come to life.

15.1 Visual Pillars

- **Cross-hatched ink illustration** for portraits, event card art, and key story moments. 1800s newspaper woodcuts, medical engravings, penny-dreadful covers. Every clone gets a portrait in this style.
- **Aged paper backgrounds** — off-white cream, subtle foxing, occasional water stains and coffee rings on framing elements. The whole UI lives on parchment.
- **Muted, restrained color** — mostly sepia, black, cream, tobacco, oxblood. Legibility is a hard requirement.
- **Hand-lettered display type** for titles (custom or IM FELL family). Clean serif for body (Playfair, Cormorant, or similar).
- **Layered paper-cut scenes** for Homestead and Town Views — several parallax-adjacent flat layers (sky, hills, buildings, fields, weather overlay) that slide subtly with time-of-day lighting.
- **Weather and lighting as a lens.** Fog, dust, rain, snow, moonlight, blood-red dusk are overlay filters. The scene does not repaint; the world gets lit differently.
- **Motion is small and specific.** A crow flies across the sky. Smoke rises from a chimney. A lantern lights up at dusk. Wheat sways. The game feels alive through the smallest possible movements.

15.2 Seasonal Color Accents

Season	Accent Color
Spring	pale moss green
Summer	wheat gold
Fall	rust orange
Winter	bone blue
Reckoning (mode)	desaturated arterial red — only when meter is high

15.3 Reference Constellation

Sunless Sea. Frostpunk's card art. 80 Days. Reigns. Banner Saga silhouettes. Illustrations in a Poe collected works. Farmer's Almanac woodcuts. Bosch's engravings.

15.4 The Look at a Glance

An 1800s almanac someone has been living out of for years. Ink illustrations of somber-eyed farmhands. Paper-cut fields under a dying yellow sun. A crow, mid-flight, across a wheat-gold sky. Something is written in the margins in a different hand.

15.5 Production Sizing (Rough)

- Illustrated portraits: ~ 40 base clone $\times$ 3 age states + ~ 20 named NPCs $\times$ 3 states $\approx$ 200 portrait assets.
- Event card illustrations: ~ 120 for MVP, growing to 400+ post-launch.
- Homestead scene layers: ~ 30 core $\times$ seasonal/weather variants $\approx$ 180 layer assets.
- Town scene: similar.
- UI system: custom illustrated frames, buttons, dividers — a full Almanac UI kit.

Total MVP scope: one senior illustrator + one supporting artist over 8-12 months, delivered alongside development.

16 UI, UX & Mobile Layout

16.1 Orientation & Layout

Portrait orientation is primary. Everything must work on a phone held one-handed. Landscape is supported as a tablet/PC courtesy, not a first-class layout.

16.2 Tab Bar & Navigation

Bottom tab bar: Homestead, Clones, Fields & Stock, Market, Town, Ledger. Six tabs, each with a distinct purpose. Icons large, spaced, labeled.

The "Advance Day / End Day" action button sits above the tab bar in a fixed location. Muscle memory. Turns red when there is an unmade decision that would be lost by advancing.

16.3 Touch Targets & Accessibility

- Minimum 44x44pt for every tappable element (Apple HIG / Android 48dp).
- Adjustable text size (Small / Standard / Large) with a slider for fine control.
- High-contrast mode toggle — thicker outlines, harsher black-on-cream.
- Colorblind-safe — no red/green as sole information carriers; every color-coded state has an icon backup.
- Haptic feedback on decision cards, harvest completions, market spikes.
- VoiceOver / TalkBack — every card readable end-to-end; every icon labeled. Planned into the framework from day one.
- No timing-based mechanics — no dexterity requirements, no reaction-time barriers.

16.4 Screen Priority (Time-of-Eyes)

- 1. Homestead View — default, ambient.
- 2. Choice Cards — high emotional weight.
- 3. Fields & Stock — mid-frequency planning.
- 4. Clones — moderate, mostly at dawn assignment.
- 5. Market — spikes hard during trade sessions.
- 6. Town — only when traveled to, but dense when active.
- 7. Ledger — occasional deep-dive, especially between runs.

Polish investment follows this ranking. Homestead View gets the most love. Ledger can be functional first, beautiful later.

17 Technical Architecture

17.1 Stack Overview

The game is turn-based, image-and-menu-heavy, low-graphics-intensity, text-rich, save-state-heavy. This profile maps cleanly to a Next.js + Capacitor stack with no game engine. We are building a text-and-illustration application that happens to be a game.

Layer	Choice
Frontend framework	Next.js 15 (App Router)
UI	React + Tailwind CSS
Animation	Framer Motion + CSS + occasional Canvas
Audio	Howler.js
State management	Zustand + Immer
Mobile packaging	Capacitor (iOS + Android from same codebase)
Hosting	Vercel (web) + EAS or native builds (stores)
Database	Neon (serverless Postgres)
Auth	Better Auth (self-hosted, TypeScript-first)
ORM	Drizzle
Content pipeline	YAML/JSON event files loaded at runtime
Testing	Vitest (unit), Playwright (E2E), headless sims (balance)
Analytics	PostHog (or Vercel Analytics)
Crash reporting	Sentry

17.2 Frontend

Next.js App Router, client-side rendered by default. Static export enabled so the same build wraps for iOS/Android with no server dependency. Server routes exist only for save sync, auth handshakes, IAP verification.

Homestead/Town dioramas are stacked `<img>` layers with CSS transforms for parallax and a single Canvas overlay for weather/lighting particles. No game engine — trivial draw cost, mobile-friendly.

17.3 State & Game Logic

The entire game state is one plain-JS object — fields, clones, market, events, reckoning, town relationships, calendar — all serializable. Save/load is `JSON.stringify`.

Game logic is pure functions: `(state, action) => newState`. Same for market ticks, crop growth, event triggers, reckoning calculations. This means:

- Unit-testable in isolation (Vitest), independent of UI.
- Headless simulations for balance — spawn 10,000 runs, check survival distribution.
- Deterministic replays — a seed + action log = exact reproduction. Critical for bug reports.

All randomness uses a seeded PRNG. Seed is stored in save state. Enables daily-seed challenges post-launch.

17.4 Save System

- **Local save (IndexedDB).** Auto-save on every day-end. 3-5 manual slots. Under 100KB serialized even in a long lineage.
- **Cloud save (Neon).** Optional, requires account. JSON blob in Postgres per slot. Last-write-wins with conflict warning if devices diverge.
- **Cross-platform sync.** Start on web, continue on iPhone, keep going on Android. Save follows a signed-in player.

17.5 Mobile Packaging (Capacitor)

Capacitor wraps the Next.js static export as native iOS/Android apps. One codebase, three platforms. Native features via plugins: haptics, push notifications, IAP, secure storage, share sheet. App size ~15-30 MB depending on how much art ships in-bundle vs. lazy-load.

PWA support in parallel for Android users who prefer to skip the Play Store. iOS App Store is the real iOS distribution story.

17.6 Content Pipeline

Events, dialogue, and story arcs live in structured YAML/JSON files loaded at runtime.

This means writers can add content without touching code. Post-launch content drops are asset updates, not app updates. Community modding is a straight path if we ever open it up.

17.7 Backend: Neon + Better Auth + Vercel

- **Database — Neon.** Serverless Postgres, native Vercel Marketplace integration, HTTP driver works in edge functions, preview branches get their own Neon branch.
- **Auth — Better Auth.** TypeScript-first, self-hosted. Handles email/password, magic links, Apple + Google sign-in. Official Drizzle adapter. Session cookies on web, JWTs for Capacitor webviews.
- **Hosting — Vercel.** Web app + serverless API routes. Preview deploys per branch. Free tier through alpha.

17.8 Database Schema (Sketch)

Small schema. JSON blobs carry game state and codex; only entitlements need real relational structure.

- users — managed by Better Auth.
- saves — id, user_id, slot, game_state (jsonb), schema_version, updated_at, device_id.
- codex_persistent — user_id, codex_state (jsonb), updated_at.
- purchases — id, user_id, platform, product_id, receipt, granted_at.

17.9 Testing Strategy

- Unit tests (Vitest) — every game logic module. Target 80%+ coverage of logic layer.
- Integration tests — headless simulated runs, 500 days of scripted actions, assert final state. Catches balance regressions.
- E2E (Playwright) — critical paths only on every PR.
- Playtesting — real humans from Alpha onward. Formal playtest program from Beta.
- Balance simulation — Monte Carlo runs, 10,000 games per strategy, chart survival rates.

17.10 Analytics & Performance

Analytics used for balancing, not surveillance. Session length, progression funnel, cause of death distribution, cruelty distribution, feature usage, retention cohorts. All opt-out-able.

Performance targets:

- Cold app open to Homestead View: <2 seconds on mid-range 2022 phone.
- Day advance (quiet): <100ms end-to-end.
- Day advance (event card): <300ms including card art load.
- Total install size: <30 MB (art delivered progressively).
- Memory footprint: <150 MB.
- Battery cost: near-zero when idle (no continuous rendering, no polling).

18 Team, Timeline & Sizing

18.1 Effective Team Shape (MVP)

- 1 full-stack developer (Next.js, game logic, Capacitor packaging).
- 1 senior illustrator (Almanac art, portraits, cards).
- 1 supporting artist (scene layers, UI kit, seasonal variants).
- 1 game designer / writer (event content, dialogue, arcs — could be the founder).
- 1 audio designer (part-time or contract — music + SFX).
- Playtesters (contract, from Beta).

18.2 Timeline (Rough)

Phase	Focus
Months 1-2	Setup — Next.js/Vercel/Neon/Better Auth wiring, project scaffolding, art bible.
Months 3-5	Vertical slice — one field, one clone, one crop, one season, one event.
Months 6-8	Alpha — all core systems implemented, ~40 events, playable through one year.
Months 9-11	Beta — art complete, content targets hit, playtesting, balance tuning.
Month 12	Polish, store submission prep, marketing site launch.
Post-launch	Content drops every 6-8 weeks (event packs, NPC threads, Season Arcs).

10-14 month MVP is the realistic target for this team shape.

19 Still To Be Designed

The following sections were not covered in the foundational brainstorming and will be added to the next revision of this document.

19.1 MVP Scope Definition

Precise cut list — which of the systems described here ship at 1.0 and which are post-launch. Recommended lens: ship the core loop (farming + market + clones + reckoning + one town + one Season Arc + ~120 events) and defer the Vat, Black Market, Rail Depot, and 4+ archetypes to post-launch content updates.

19.2 Monetization Model

Four options on the table:

- Premium paid (\$4-8, no IAP, no ads) — cleanest, best reviews, hardest to market.
- Free demo + full-game unlock (\$4-8) — good conversion, no dark patterns.
- F2P with cosmetics + DLC expansion packs.
- Full F2P with ads and IAP — highest ceiling, worst design constraints. Not recommended.

Requires a business-model decision before store submission planning.

19.3 Marketing & Community Strategy

Launch positioning, press outreach, streamer strategy, Discord community, wishlist campaign on itch/Steam if PC version considered.

19.4 Localization Plan

The text-heavy content pipeline is localization-friendly. Priority languages, timing (launch vs. post-launch), and translation vendor decisions.

19.5 Live-Ops & Post-Launch Content Cadence

Rough plan: event pack every 6-8 weeks, major content drop (new Season Arc or new NPC) every quarter, expansion (new land region + archetype + weird crop) yearly.

A Appendix A: Glossary

Ascension. Optional difficulty modifiers unlocked in meta-progression (+1 through +10).

Bone-root. Weird crop; grows better in fields where a clone has died. Sells only at the Black Market.

Bushel. Unit of yield for grain and roots.

Codex. Persistent memorial book that tracks every farmer, clone, year, record, and lore fragment across all runs.

Cruelty Ledger. Four-axis tracking of moral posture: Reckoning Meter, Household Morale, Town Reputation, Ghost Roll.

Dr. Ambrose Vane. Traveling clone merchant. Brother of Meredith Vane.

Family Plot. A dedicated burial ground on the farm; funerals held here reduce reckoning more effectively.

Foundling. A clone who arrives at the gate from elsewhere; unknown quality, unknown history.

Ghost Roll. Hidden list of wronged dead. The reckoning draws from it when it walks.

Homestead View. The default screen — a 2D diorama of the farm.

Legacy Report. The novel-like ending sequence at run-end.

Line-death. End condition: no successor available and no coin to hire one.

Marrow's Cross. The town.

Meredith Vane. Saloonkeeper. Sister of Dr. Ambrose Vane.

Moon Barley. Weird crop; grows only if planted under a full moon.

Ossuary. Late-game building; keeps clone bones on site with mechanical benefits and terrible costs.

Reckoning. The supernatural response to accumulated cruelty. Escalates through five tiers.

Season Arc. One town-wide story per in-game year.

The Long Vigil. A midsummer church rite. Attendance reduces reckoning.

The Vat. Mid-game building for growing clones from raw material.

Vigils. The meta-currency earned at run-end and spent on unlocks.

Walker. A dead clone who returns at night with a specific grievance.

Whisper Wheat. Weird crop; whispers to clones if left unharvested at night.

B Appendix B: NPC Reference Table

Name	Role	Function
Mayor Cyrus Halloway	Politics	Civic contracts, town-wide policies.
Preacher Elias Grange	Church	Blesses or condemns publicly. Moral weathervane.
Doc Bell	Medicine	Medicine, rumors. Cynical, sees all.
Meredith Vane	Saloon	Information hub. Sister of Ambrose.
Silas Ridley	Bank	Loans, mortgages, foreclosure countdowns.
Bess Halloway	Marriage	Potential spouse/heir source.
Old Nan	Folk Magic	Reckoning intel. Advance warnings.
Sheriff Nathaniel Coldwater	Law	Cruelty investigations. Fines, seizures.
Dr. Ambrose Vane	Clone Merchant	Sells clones from a traveling wagon.
Sister Ruth Grange	Charity	Donations boost morale, reduce reckoning.

C Appendix C: Foundational Decisions Log

This log captures the sequence of design decisions made during the foundational brainstorming session, with rationale, for future reference.

Core Player Fantasy: "I survived another year." Grit, scarcity, hard-won victories against nature. Chosen over legacy/cunning/whimsy fantasies to set the tonal spine.

Setting: Alternate 1800s with darker tone and light story. Multiple gameplay pillars — not just market/trading. Pitched as Stardew Valley × Oregon Trail × ethical horror.

Weirdness Dial: Weird West / occult undertow. Something is genuinely wrong out there. Slow-burn horror over overt fantasy.

Core Hook: Player buys farmer clones using crazy 1800s pseudo-science. Player can treat them well or cruelly; cruelty raises yields but calls consequences.

Consequences: Supernatural — the land remembers. Hauntings, blights, ill omens, the dead do not stay buried. Tightest thematic fit.

Spatial Frame: Rooted homestead + a nearby town (Stardew-shape). Oregon Trail DNA relocates from travel to weighty menu decisions, brutal events, and resource management.

Run Structure: Nested: Year → Farmer's Lifetime → Family Lineage on Land. Hybrid of tight tactical loop and long generational arc. Runs end only on line-death or land-loss.

Time Model: Turn-based days, 20-day seasons, 80-day years. Portrait mobile-first. Session sweet spot 10-15 minutes.

Day Structure: Three-beat day: Morning Brief → Play → Dusk Report. Homestead View as static diorama. Per-clone assignment with auto-templates for late-game scaling.

Farming Systems: Four crop families (Grains, Roots, Cash, Weird). Field-level abstraction (not tile-level). Named fields with history.

Clones: Three sources (Merchant, Vat, Foundlings). Four-axis cruelty ledger. Five burial options.

Market: Four price layers, four venues, forward contracts, layered information sources.

Town: Marrow's Cross. Ten named NPCs. Season Arcs + NPC threads. Four annual festivals with soft-required attendance.

Meta-Progression: Vigils meta-currency. Legacy Report as novel-ending. Codex persists across all runs. Marrow's Cross evolves. Ascension modifiers (+1 to +10).

Events: Seven event families. Choice-card resolution. Five Reckoning tiers. ~120 events at MVP.

Art: The Illustrated Almanac — cross-hatched ink, aged paper, seasonal accents.

UI: Portrait-primary, six-tab bottom bar. Full accessibility from day one.

Tech: Next.js + Capacitor, no game engine. Neon + Better Auth + Drizzle + Vercel.
Content in YAML.

End of Document
Bushel & Bone — GDD v0.1